

General model of microtubules

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Abstract In the present work, we deal with nonlinear dynamics of microtubules. A new model, describing nonlinear dynamics of microtubules, is introduced. Its advantages over two existing models are demonstrated. We show that dynamics of microtubules can be explained in terms of kink solitons. Also, circumstances yielding to either subsonic or supersonic solitons are discussed.

Keywords Microtubules · Kink solitons · General model

1 Introduction

Microtubules (MTs) are the basic components of cytoskeleton of eukaryotic cells and play essential role in the shaping and the maintenance of cells. They are involved in cell division and in a variety of cellular

traffic processes together with motor proteins, primarily kinesin and dynein [1]. In that respect, MTs are not just sturdy tracks for motor advances but they also exhibit their own dynamical changes in this process.

The basic unit of a MT is $\alpha\beta$ -tubulin heterodimer protein. These dimers polymerize in straight protofilaments (PFs), which aggregate laterally into cylindrical tubular structure. For the model introduced in this paper, the dimers are assumed to be elementary particles whose length is $l = 8$ nm. They perform angular oscillations around the axes positioned through their centres of mass.

These dimers arrange head to tail forming biologically polar MTs with a more dynamic plus end and less active minus end. Important for this article is the fact that tubulin dimers represent electric dipoles [2–4] and, consequently, MT appears to be a giant dipole as a whole with negatively charged end coinciding with biologically positive end and vice versa. This is the reason why within a MT an intrinsic electric field exists.

These circumstances were incorporated in the framework of the model considered in this paper. Both α - and β -tubulin bind one GTP molecule but only GTP within β -tubulin is being hydrolyzed after polymerization. This chemical event leads to remarkable conformational change within the dimer which can trigger the launching of a kink wave along pertaining protofilament of MT.

It was demonstrated that the interaction between progressing kinesin motor and tubulin dimers within pertaining MT leads to appearance of profound con-

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formations of α - and β -tubulin [5]. These conformational displacements can also be considered as one of the mechanisms responsible for creation of kink-waves that are one of the most promising dynamical mechanisms for intracellular signalling.

There are a few models describing nonlinear dynamics of MTs. Beside some attempts, all of them assume one degree of freedom per dimer. The first of these models, relevant for this paper, is a longitudinal one [2]. In fact, the dimers perform angular oscillations but a coordinate u , describing the dimers displacement, is a projection of the top of the dimer on the direction of PF. Hence, this is the longitudinal coordinate and the model represents the longitudinal one. There is a real longitudinal model, assuming the longitudinal displacements [3], but this is not relevant for this paper. The model is based on the fact that the dimers are electric dipoles and, consequently, the whole MT can be regarded as ferroelectric [2,4]. Somewhat improved version of the mentioned model is what we have called as u - model [6]. A more precise definition of the coordinate u can be found in Ref. [7].

There exists an angular model as well that we have called as φ - model [8]. The coordinate φ is the angle between directions of the dimer and PF. A purpose of this paper is to introduce a new model that we call as general model (GM), which overcomes some weak points of the models mentioned above. Hence, in what follows, we explain very briefly both u and φ - models and introduce the new one. We show its advantages and close the paper with some concluding remarks.

2 General model of microtubules

Before we introduce the new model that we call as the general model we very briefly explain the existing u and φ - models. There exist two types of contacts between the dimers. These are head-to-tail binding of dimers, along PFs, and interactions between parallel PFs, which form the MT wall. It is known that the longitudinal contacts along PFs are much stronger than those between adjacent PFs [9,10]. This means that Hamiltonian of MT dynamics is, practically, Hamiltonian for a single PF only. However, the influence of the neighbouring PFs is taken into consideration through the electric field. Namely, each dimer exists in the electric field coming from the dimers belonging to all PFs. Also, the nearest neighbour approximation is safely assumed.

The Hamiltonian for the u - model is [2,6,7]

$$H_u = \sum_n \left[\frac{m}{2} \dot{u}_n^2 + \frac{k_u}{2} (u_{n+1} - u_n)^2 - \frac{1}{2} A_u u_n^2 + \frac{1}{4} B_u u_n^4 - C u_n \right], \quad C = QE, \quad (1)$$

where dot means the first derivative with respect to time, m is a mass of the single dimer, k_u is an inter-dimer stiffness parameter, $Q > 0$ represents the excess charge within the dipole, $E > 0$ is intrinsic electric field, and the integer n determines the position of the considered dimer in PF. The first two terms represent the kinetic energy of the dimer and the potential energy of the chemical interaction between the neighbouring dimers belonging to the same PF, respectively. The next two terms represent W -potential, which was introduced due to the fact that MT is ferroelectric, where A_u and B_u are positive parameters that should be determined or, at least, estimated [2,6,7]. The very last term is coming from the fact that the dimer is the electric dipole existing in the field of all other dimers. The last three terms together can be seen as more realistic asymmetrical W -potential.

Equation (1) brings about the following partial differential equation (PDE) [2,6]

$$m \frac{\partial^2 u}{\partial t^2} - k_u l^2 \frac{\partial^2 u}{\partial x^2} - qE - A_u u + B_u u^3 + \gamma \frac{\partial u}{\partial t} = 0, \quad (2)$$

where, of course, a continuum approximation has been readily used. The last term is introduced viscosity force $F_v = -\gamma du/dt$, where γ is viscosity coefficient [2]. It is well known that Eq. (2) can be transformed into an ordinary differential equation (ODE)

$$\alpha_u \psi_u'' - \rho_u \psi_u' - \psi_u + \psi_u^3 - \sigma = 0, \quad (3)$$

where a dimensionless function ψ_u was introduced through $u = \sqrt{A_u/B_u} \psi_u$. This is a travelling wave solution $u(x, t) = u(\xi)$, depending upon x and t only through a unified variable

$$\xi = \kappa x - \omega t, \quad (4)$$

where κ and ω are constants and $\psi_u' \equiv d\psi_u/d\xi$. The expressions for the parameters existing in Eq. (3) are [6].

$$\alpha_u = \frac{m\omega^2 - k_u l^2 \kappa^2}{A_u}, \quad \sigma = \frac{qE}{A_u \sqrt{A_u/B_u}}, \quad \rho_u = \frac{\gamma\omega}{A_u}. \quad (5)$$

Notice that the crucial parameter α_u exhibits the competition between the inertial term and the elastic one in Eq. (2).

The Hamiltonian for one PF for the φ -model is [8]

$$H_\varphi = \sum_n \left[\frac{I}{2} \dot{\varphi}_n^2 + \frac{k}{2} (\varphi_{n+1} - \varphi_n)^2 - pE \cos \varphi_n \right], \quad (6)$$

where I is a moment of inertia of the single dimer and p stands for electric dipole moment. The equations corresponding to Eqs. (2) and (3) are [8]

$$I \frac{\partial^2 \varphi}{\partial t^2} - kl^2 \frac{\partial^2 \varphi}{\partial x^2} + pE \varphi - \frac{pE}{6} \varphi^3 + \Gamma \frac{\partial \varphi}{\partial t} = 0 \quad (7)$$

and

$$\alpha_\varphi \psi_\varphi'' - \rho_\varphi \psi_\varphi' + \psi_\varphi - \psi_\varphi^3 = 0, \quad (8)$$

where the function ψ_φ has been introduced through $\varphi = \psi_\varphi \sqrt{6}$ and

$$\alpha_\varphi = \frac{I\omega^2 - kl^2\kappa^2}{pE}, \quad \rho_\varphi = \frac{\omega\Gamma}{pE}. \quad (9)$$

Of course, the last term in Eq. (7) is viscosity momentum $M_v = -\Gamma\dot{\varphi}$, where Γ stands for the appropriate viscosity coefficient.

The solutions of Eqs. (3) and (8) are kink and antikink solitons. They may be considered as possible candidates for information carriers along MTs. It is important to point out that ρ_u , σ and ρ_φ are known parameters, while α_u and α_φ can be determined. It was shown that $\alpha_u < 0$ [6] and $\alpha_\varphi > 0$ [8], i.e. $m\omega^2 < k_u l^2 \kappa^2$ and $I\omega^2 > kl^2 \kappa^2$.

Before we proceed we should discuss both models. A weak point of the u -model is the last term in Eq. (1). A scalar product $-\vec{p} \cdot \vec{E} = -QdE \cos \varphi_n$ would be better choice for the potential energy, where d is the distance between centres of positive and negative charges within the dipole. This potential exists in the φ -model but the W -potential has been missing. In fact, a series expansion of $\cos \varphi_n$ gives φ_n^2 and φ_n^4 terms but with opposite signs from those in the u -model as A_u and B_u are positive parameters, which was stated above. These two terms are, practically, a potential that looks like W in a mirror having only one minimum

surrounded by two maxima and, due to its shape, can be called as M -potential. This potential brings about $\alpha_\varphi > 0$, which is disputable result. Also, we want to construct the model using the coordinate φ , which is certainly more natural than the longitudinal one since the dimers perform radial oscillations.

Therefore, we want to solve the mentioned problem regarding the u -model but to keep the W -potential, the coordinate φ and, probably, $I\omega^2 < kl^2 \kappa^2$. This suggests the following Hamiltonian

$$H = \sum_n \left[\frac{I}{2} \dot{\varphi}_n^2 + \frac{k}{2} (\varphi_{n+1} - \varphi_n)^2 - \frac{A}{2} \varphi_n^2 + \frac{B}{4} \varphi_n^4 - pE \cos \varphi_n \right], \quad (10)$$

where $A > 0$, $B > 0$ and φ_n has the same meaning as in the φ -model. The procedure mentioned above brings about

$$I \frac{\partial^2 \varphi}{\partial t^2} - kl^2 \frac{\partial^2 \varphi}{\partial x^2} - (A - pE) \varphi + \left(B - \frac{pE}{6} \right) \varphi^3 + \Gamma \frac{\partial \varphi}{\partial t} = 0 \quad (11)$$

and further

$$\left(I\omega^2 - kl^2 \kappa^2 \right) \varphi'' - \Gamma \omega \varphi' - (A - pE) \varphi + \left(B - \frac{pE}{6} \right) \varphi^3 = 0, \quad (12)$$

where, of course, $\varphi \equiv \varphi(\xi)$. Now, we need to study a few cases.

Case I We assume $(A - pE) \left(B - \frac{pE}{6} \right) > 0$, which straightforwardly yields to

$$\alpha \psi'' - \rho \psi' + \psi - \psi^3 = 0, \quad (13)$$

where

$$\varphi = \pm \sqrt{\frac{A - pE}{B - pE/6}} \psi \equiv K \psi \quad (14)$$

and

$$\alpha = \frac{I\omega^2 - kl^2 \kappa^2}{pE - A}, \quad \rho = \frac{\Gamma \omega}{pE - A}. \quad (15)$$

Formally, Eqs. (8) and (13) are equal and the solution of the latter one can be written according to the φ -model, that is [8]

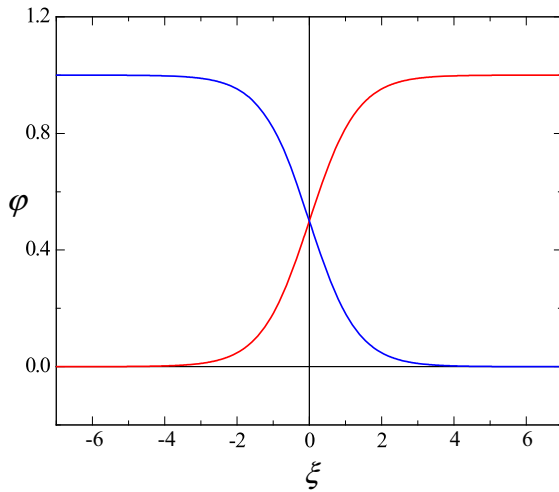


Fig. 1 Kink soliton (blue, $\rho = -1$) and antikink soliton (red, $\rho = 1$) for $K = 1$. (Color figure online)

$$\varphi(\xi) = \frac{K}{2} \left[1 + \tanh \left(\frac{3}{4\rho} \xi \right) \right], \quad \alpha > 0. \quad (16)$$

This solution differs from the one characterizing the φ -model in two points. The parameters ρ_φ and ρ are different and K stands instead of $\pm\sqrt{6}$. This will be discussed later. Notice that Eq. (16) holds for both positive and negative ρ . Hence, $\varphi(\xi)$ represents kink soliton if $\rho > 0$ and antikink one for negative ρ , which is shown in Fig. 1.

Of course, the solution (16) was obtained for $\rho \neq 0$. For the special case of negligible viscosity, one easily obtains

$$\varphi_0 = K \tanh(\xi/a), \quad (17)$$

where a is an arbitrary constant and $\alpha = a^2/2$.

Case 2 We assume $(A - pE) \left(B - \frac{pE}{6} \right) < 0$, which straightforwardly yields to

$$\alpha\psi'' - \rho\psi' + \psi + \psi^3 = 0, \quad (18)$$

where

$$\varphi = \pm \sqrt{\frac{pE - A}{B - pE/6}} \psi \equiv K'\psi. \quad (19)$$

There are a variety of methods that can be used to solve Eq. (18). As the purpose of this paper is the introduction of the new model rather than a tedious mathematical

analysis, we pick up one of the simplest. This is modified extended tanh-function method [6, 11–14], and we assume that the procedure is known. We expect the solution ψ as

$$\psi = a_0 + a\Phi, \quad (20)$$

where Φ is the solution of the well-known Riccati equation

$$\Phi' = b + \Phi^2 \quad (21)$$

and the parameters a_0 , a and b should be determined. Following, for example, Ref. [6], i.e. using Eqs. (18), (20) and (21), we straightforwardly obtain a crucial equation

$$2\alpha ab\Phi + 2\alpha a\Phi^3 - \rho ab - \rho a\Phi^2 + a_0 + a\Phi + a_0^3 + 3a_0^2a\Phi + 3a_0a^2\Phi^2 + a^3\Phi^3 = 0. \quad (22)$$

As the coefficients for Φ^3 , Φ^2 , Φ and $\Phi^0 = 1$ should simultaneously be equal to zero, we obtain a system of four equations whose solution is

$$\alpha = \frac{2\rho^2}{9} > 0, \quad b = -\frac{9}{16\rho^2}, \quad a_0^2 = -\frac{1}{4}, \quad 3a_0a = \rho. \quad (23)$$

Obviously, real solutions having physical sense do not exist. Two solutions that may be of mathematical interest are

$$\varphi_2 = i \frac{K'}{2} \left[1 + \tanh \left(\frac{3}{4\rho} \xi \right) \right] \quad (24)$$

and they are obtained assuming $\rho \neq 0$. For the special case of negligible viscosity, one easily obtains

$$\varphi_{20} = K' \tan(\xi/a), \quad (25)$$

where a is an arbitrary constant related to α through $a^2 = -2\alpha$.

Case 3 We assume $A = pE$ and $B \neq \frac{pE}{6}$, which straightforwardly yields to

$$\alpha_3\psi'' - \rho_3\psi' + \psi^3 = 0, \quad (26)$$

where

$$\alpha_3 = \frac{I\omega^2 - kl^2\kappa^2}{B - pE/6}, \quad \rho_3 = \frac{\Gamma\omega}{B - pE/6}. \quad (27)$$

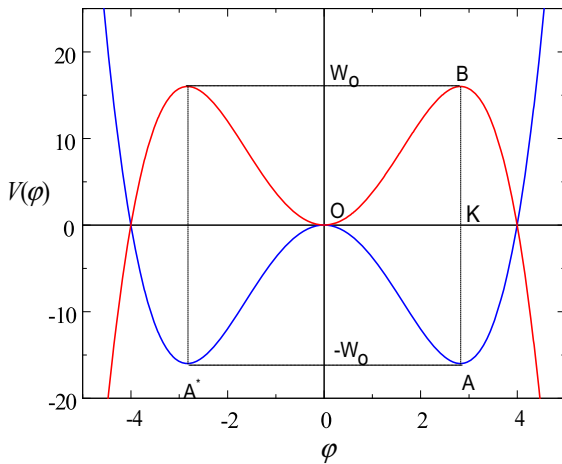


Fig. 2 Functions $V(\varphi) = -C_1\varphi^2 + C_2\varphi^4$ for $C_1 = 4$, $C_2 = 0.25$ (blue) and $C_1 = -4$, $C_2 = -0.25$ (red). (Color figure online)

The procedure used for Case 2 brings about the equation equal to Eq. (22) except that the terms a_0 and $a\Phi$ do not exist and, of course, α_3 and ρ_3 replace α and ρ in Eq. (22). Its solution is $a_0 = a = \alpha_3 = \rho_3 = 0$, which certainly means that Eq. (26) does not have any solution having physical sense.

The remaining case $B = pE/6$ linearizes Eq. (12) and will not be studied here.

3 Discussions

We point out some advantages of GM. First of all, GM uses the angle φ as the coordinate, which is certainly more natural and convenient than the longitudinal one existing in the u -model. Also, the problem with the last term in Eq. (1) has been solved as explained above.

Let us discuss more important points. It was explained above that the u -model is characterized by the W -potential, while the series expansion of the cosine function yields to the M -potential. This is shown in Fig. 2. The corresponding potential energy in GM is

$$V(\varphi) = -\frac{1}{2}(A - pE)\varphi^2 + \frac{1}{4}\left(B - \frac{pE}{6}\right)\varphi^4, \quad (28)$$

which comes from Eq. (1). Also, the numerator in Eq. (15) can be represented as

$$I\omega^2 - kl^2\kappa^2 = I\kappa^2 \left[\left(\frac{\omega}{\kappa}\right)^2 - \frac{kl^2}{I} \right] \equiv I\kappa^2 (v^2 - c^2), \quad (29)$$

where v and c are the soliton and the sound velocity, respectively. Hence, Eqs. (15), (16), (28) and (29) yield to the final conclusion:

- If $A > pE$ and $B > pE/6$, then $\rho < 0$, $V(\varphi)$ is W -potential and the function $\varphi(x, t)$ is subsonic soliton, kink for the positive K in Eq. (16) and antikink otherwise.
- If $A < pE$ and $B < pE/6$, then $\rho > 0$, $V(\varphi)$ is M -potential and the function $\varphi(x, t)$ is supersonic soliton, antikink for the positive K in Eq. (16) and kink otherwise.

To completely understand physical meaning of the mentioned solitons, we should see asymptotic values of $\varphi(\xi)$ given by Eq. (16) for both cases, that is:

$$a) \varphi_a(-\infty) = K, \varphi_a(+\infty) = 0 \quad (30a)$$

$$b) \varphi_b(-\infty) = 0, \varphi_b(+\infty) = K. \quad (30b)$$

This means, as shown in Fig. 2, that φ_a and φ_b represent jumps from A to 0 and from 0 to B , respectively. When the system is in 0, it will spontaneously move to one of the potential wells, A or A^* . However, the system in B may move to 0 but may not. This means that only the case a corresponds to the stable wave solution of Eq. (12).

Let us explain one more problem within the φ -model solved by our GM. The amplitude of φ for the φ -model is $\sqrt{6}$ [8]. This value comes from the series expansion of the cosine function and has nothing to do with any physical argument. Also, this value is very big, about 140° , and it is extremely unlikely that the dimers perform so big angular displacements during soliton transmission. As for GM, this angle is K , given by Eq. (14). Its possible value will be estimated in what follows.

4 Estimations

We need to estimate the values of the parameters A and B , existing in Eq. (10). Let us concentrate on the blue line in Fig. 2. Suppose that the value of K is a known angle φ_1 , i.e.

$$\sqrt{\frac{A - pE}{B - pE/6}} = \varphi_1. \quad (31)$$

Suppose, also, that W_0 is known, that is

$$-V(\pm K) = \frac{1}{4} \frac{(A - pE)^2}{B - pE/6} = W_0 > 0. \quad (32)$$

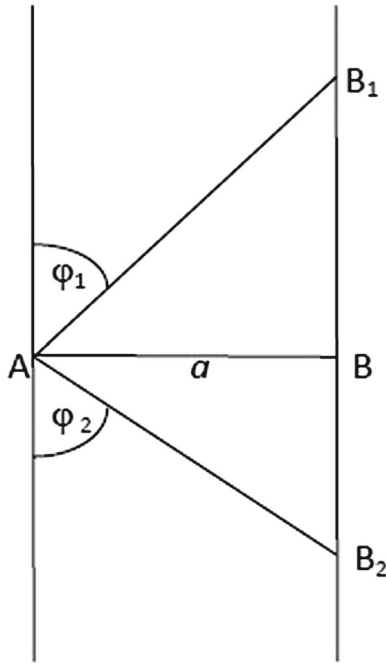


Fig. 3 Segments of two neighbouring PFs

From Eqs. (31) and (32), we easily obtain

$$A = pE + \frac{4W_0}{\varphi_1^2}, B = \frac{pE}{6} + \frac{4W_0}{\varphi_1^4}. \quad (33)$$

It is known that supply of energy from hydrolysis of guanosine triphosphate (GTP) in MTs may excite the vibrations [15]. Hence, W_0 can be identified with this energy, i.e. $W_0 = 0.25$ eV. Also, we use: $p = 337$ Debye $= 1.13 \times 10^{-27}$ cm [16, 17] and $E = 1.7 \times 10^7$ N/C [3]. To estimate the angle φ_1 , we use Fig. 3, representing a short segment of two neighbouring PFs. The points A, B₁ and B₂ are centres of the three dimers and $\overline{B_1 B_2} = l$. It is known that the dimers in the neighbouring PFs are shifted for $8l/13$ [18, 19]. Hence, $\overline{BB_1} = 8l/13$ and $\overline{BB_2} = 5l/13$. Also, $a = 0.6l$ [3]. Orientations of the dimers are not known. According to MT geometry, it is extremely unlikely that the dimers are in the direction of PFs. The directions AB₁ and AB₂ are certainly better choices. If we assume that after the energy release the dimer does not exceed the PF direction we can state that the displacement is not bigger than φ_1 . As shown in Fig. 3, we easily calculate $\varphi_1 = 0.77$ rad, $\varphi_1^2 \approx 0.6$ and $\varphi_2 = 1$ rad, which, according to Eq. (33), brings about $A = 1.8$ eV and $B = 2.8$ eV. Therefore, Fig. 1 was done for $K = 1$.

More precise values are φ_1 or φ_2 . Hence, instead of too big value of about 140° the amplitude is about 40° , which makes sense. A patient reader might have noticed that the angles φ_1 or φ_2 , corresponding to the points A and A* in Fig. 2, are not equal, while the W -potential is symmetric. This is correct, and Fig. 2 is somewhat simplified. The next step in research of nonlinear dynamics of MTs will be introduction of more realistic asymmetric W -potential. This can be done introducing one more term in Eq. (10). Hence, the potential should be either $-\frac{A}{2}\varphi_n^2 + \frac{B}{4}\varphi_n^4 + C\varphi_n$ or $-\frac{A}{2}\varphi_n^2 + \frac{B}{4}\varphi_n^4 + C\varphi_n^3$, where C is a new constant to be determined, but this is beyond this paper. We could contemplate that one of the reasons why MT lattice is helical is the introduction of symmetry breaking [20], which leads to proper asymmetric shape of W -potential in the context of here presented GM of MT dynamics.

Finally, we want to relate the soliton velocity with the most intriguing parameter Γ . From Eqs. (4) and (16) we easily obtain

$$\frac{3\kappa}{4|\rho|} = \frac{2\pi}{\Lambda} = \frac{2\pi}{Nl}, \quad (34)$$

where Λ is the soliton width and N is an integer. Using Eq. (15) we obtain

$$v = \frac{3Nl|A - pE|}{8\pi\Gamma}. \quad (35)$$

Equation (35) can be used for rough estimation of the parameter Γ . There should be $v < c$, where $c = 340$ m/s is sound speed. Using Eqs. (33) and (35), as well as the estimated values for W_0 and φ_1^2 , we easily obtain

$$\Gamma > 0.75 N \times 10^{-30} \text{ Nms}. \quad (36)$$

Now, we can estimate the kink's velocity on the basis of the above-calculated parameters. The key point is the order of magnitude of viscous damping parameter Γ . This parameter can be calculated relying on the calculation presented in Ref. [21]. As we consider the dimer as a cylinder of length l performing angular oscillations around the axis orthogonal to its centre of mass, we get the expression

$$\Gamma = \frac{1}{12} \eta l^3, \quad (37)$$

where η stands for effective viscosity. Taking that the number of dimers N involved in a kink is of the order of ten and using the viscosity coefficient for water, which is 10^{-3} Ns/m² at room temperature, we obtain $\Gamma \approx 43 \times 10^{-30}$ Nms. According to Eq. (35), this yields to $v \approx 60$ m/s. Notice that the assumed value for N and the calculated value for Γ are in agreement with the requirement (36). However, this is much overestimated value. Namely, MT itself makes viscoelastic environment for the motion of displaced dimers so the effective friction coefficient Γ_{eff} can be by many order of magnitude greater compared with clear water. It was shown [22,23] that, for typical globular protein of radius of 2.5 nm, the enhancement factor appears to be as large as 750. With such circumstance taken into account, we obtain the order of magnitude

$$v \approx 0.08 \frac{\text{m}}{\text{s}} = 8 \frac{\text{cm}}{\text{s}}, \quad (38)$$

which is more realistic result comparable with the solitonic speed presented in Ref. [24].

5 Conclusion and future research

In this paper, we pointed out some weak points of the previous models of MTs. These problems have been solved introducing the GM.

Regarding future research, our theory should be related to some experimental results. It is very likely that orientation of dimers, i.e. the values of φ_1 and φ_2 , could be experimentally determined. This would improve our theory a lot.

It was pointed out that Eq. (2) was obtained from Eq. (1) using continuum approximation. Validity of this approximation was approved in Ref. [25]. However, semi-discrete approximation should also be used as was done within the u -model [7] and the φ -model [26]. It is interesting that the final solution depends not only on the physical system but on the applied mathematical procedure. Namely, instead of kinks, semi-discrete approximation yields completely different kind of solitons. These are localized modulated solitons called breathers. Therefore, semi-discrete approximation should be performed within the GM.

The interaction between the neighbouring dimers was modelled by the harmonic potential energy in the Hamiltonian (1). But, recently suggested idea to use

more realistic Morse potential instead of the harmonic one in Eq. (1) [27] should be performed within the GM.

Finally, let us come back to continuum approximation, used in this paper. To solve Eqs. (13), (18) and (26), we picked up the modified extended tanh-function method because of its simplicity. Regarding this paper, this should be enough as the purpose of this work is qualitative introduction of the new model rather than series mathematical analysis. However, different procedures, such as the simplest equation method [28–30], may bring about new solutions and should be performed.

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